SURVEYING & LEVELLING · NOTE SL16

On-Farm Water Distribution Layout — Open Channels and Underground Pipelines

Course topics: IDE5.5 L7, IDE5.5 L8, CAE1.3 L13, IDE-503 T9

Two kinds of system

Open channel system	Underground pipeline system
Common in canal-irrigated areas; flow by gravity	Water flows under pressure
Lay channels along ridge lines; on sloping land run them along the higher reach and level the downfield area to a non-erosive grade	Lay pipes along the most convenient, shortest route, with outlets at commanding elevations

Why underground pipelines

- Water reaches every field and is available close by.
- No seepage, evaporation or breach losses, and no farm land lost to channels.
- Uniform discharge from the outlets; freedom of alignment and flexible outlet location.
- Low maintenance cost and full control of the water supply.
- **Pipes:** RCC or PVC — PVC is preferred because it is easy to lay, has leak-proof joints and resists pressure.

Water-course command areas

- A **water course** starts at a canal outlet, serves a group of farmers and is maintained by them; add branch water courses where fields cannot be served directly.
- **Rectangular (consolidated) holdings** — common in Punjab, Haryana and western Uttar Pradesh — follow **warabandi**: each farmer's share of water is proportional to his land holding.
- **Unconsolidated holdings** have irregular boundaries, so plan the outlet locations carefully on contour maps.

1. Lay the water course along the ridge line, preferably along field boundaries.
2. Lay drainage channels along the valley lines.
3. Provide branch water courses where needed.

Pipelines in deep-tubewell command areas

Layout	Where and how
Branching network	Undulating or steep areas; only the main line and the branch in use carry flow; pipeline cost is about 50% higher than in a looped system
Looped network	Plain areas; both arms of a loop share the flow, so pipes can be about 30% smaller in diameter

Design requirements

1. Prepare a contour map at about 30 cm interval on the village cadastral ("chadar") map, showing field boundaries, streams, tanks, roads, wells, and the tubewell and elevated-tank locations.
2. Fix the alignment from the field contours, ground features, valve locations and field structures.

Sources: ICAR eCourse 'Irrigation Engineering' — Dr K. N. Tiwari and Dr N. S. Raghuwanshi (IIT Kharagpur), reviewed by Dr S. D. Gorantiwar (MPKV Rahuri), Lessons 17–18 (land levelling design). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. In open-channel systems, water courses are laid along

(a) valley lines
(b) ridge lines
(c) field centres
(d) drains

Q2. Warabandi allots canal water in proportion to

(a) crop type
(b) land holding
(c) distance from outlet
(d) family size

Q3. A looped pipeline network allows pipe diameters smaller by about

(a) 10%
(b) 30%
(c) 50%
(d) 70%

Q4. Pipe preferred for underground distribution because of leak-proof joints and pressure resistance

(a) RCC
(b) PVC
(c) cast iron
(d) asbestos cement

Q5. Contour interval for designing a deep-tubewell pipeline network is about

(a) 5 cm
(b) 30 cm
(c) 1 m
(d) 5 m

Answers

Q1 — (b) ridge lines
Q2 — (b) land holding
Q3 — (b) 30%
Q4 — (b) PVC
Q5 — (b) 30 cm